M15 Review / Repair - Teaching RC2

OPEN ONLY AFTER SAVED ATTEMPT

P01 repair - Cloud fundamentals for data engineers

Cloud moves responsibility boundaries; it does not make architecture/quality/security disappear.

Repair principle: Cloud provider operates physical infrastructure and managed service layers according to service model; customer still owns data/config/identity/workload correctness.

Evidence focus: cost -> usage/capacity responsibility

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P02 repair - Identity, IAM and least privilege

Cloud data platforms centralize valuable data; an over-privileged identity can expose or modify far more than one local file.

Repair principle: Authentication proves identity; authorization decides permitted actions.

Evidence focus: production admin -> limited trusted operators

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P03 repair - Azure Storage concepts and OneLake

Fabric items often store/access data through OneLake. Paths and ownership still matter for lineage, duplication and security.

Repair principle: Cloud object/lake storage organizes data by logical paths/objects, not local disk assumptions.

Evidence focus: shortcut -> logical reference to external/other data

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P04 repair - Fabric workspace and item model

Workspace/item boundaries influence deployment, access, ownership and environment separation.

Repair principle: A workspace groups Fabric items and is a collaboration/security/deployment boundary.

Evidence focus: Warehouse/SQL endpoint -> SQL serving depending need

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P05 repair - OneLake paths, files, tables and shortcuts

The same data can be physically copied, registered as table or referenced. Wrong choice causes duplication/staleness or missing transactional semantics.

Repair principle: Files are appropriate for raw/landing/artifacts without table semantics.

Evidence focus: table -> structured transactional/query semantics

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P06 repair - Fabric Lakehouse and Delta tables

Fabric simplifies lakehouse management but Delta grain/schema/MERGE/replay rules remain your responsibility.

Repair principle: Fabric Lakehouse combines Files and managed Tables backed by OneLake/Delta.

Evidence focus: SQL endpoint -> query/serving access, not quality guarantee

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P07 repair - Fabric notebooks

Cloud notebooks are convenient but hidden cell state/manual ordering can create results nobody can reproduce.

Repair principle: Keep setup/config/input parameters near the top and avoid hidden interactive state.

Evidence focus: outputs -> durable table + controls

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P08 repair - Spark in Fabric

Managed Spark removes cluster setup burden but not shuffle/skew/collect/partition mistakes or compute cost.

Repair principle: SparkSession/DataFrame concepts remain the same.

Evidence focus: session lifecycle -> capacity use

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P09 repair - Fabric Data Pipelines

Cloud pipelines coordinate movement/notebooks/SQL activities but can still publish partial or duplicate data if state/idempotency is weak.

Repair principle: Pipeline activities can copy data, invoke notebooks/stored procedures and coordinate dependencies depending service features.

Evidence focus: publish/state -> idempotent after validation

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P10 repair - SQL analytics endpoint and warehouse choice

Fabric offers more than one SQL-facing experience. Choosing by name instead of requirement can create awkward write/governance/performance patterns.

Repair principle: Lakehouse SQL analytics endpoint exposes SQL query access over Lakehouse tables; exact capabilities evolve and should be checked in current product docs.

Evidence focus: same data in both -> justify duplication/SLA

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.